



Assessed Coursework

Course Name	Text-as-Data			
Coursework Number	1			
Deadline	Time:	4:30pm	Date:	19th March 2026
% Contribution to final course mark	18			
Solo or Group ✓	Solo	✓	Group	
Anticipated Hours	20 hours			
Submission Instructions	As per specification below.			
Please Note: This Coursework cannot be Re-Assessed				

Code of Assessment Rules for Coursework Submission

Deadlines for the submission of coursework which is to be formally assessed will be published in course documentation, and work which is submitted later than the deadline will be subject to penalty as set out below.

The primary grade and secondary band awarded for coursework which is submitted after the published deadline will be calculated as follows:

- (i) in respect of work submitted not more than five working days after the deadline
 - a. the work will be assessed in the usual way;
 - b. the primary grade and secondary band so determined will then be reduced by two secondary bands for each working day (or part of a working day) the work was submitted late.
- (ii) work submitted more than five working days after the deadline will be awarded Grade H.

Penalties for late submission of coursework will not be imposed if good cause is established for the late submission. You should submit documents supporting good cause via MyCampus.

Penalty for non-adherence to Submission Instructions is 2 bands

Text-as-Data Coursework

Introduction

You have been hired as a 'fixer' for a company that develops bespoke machine learning models to classify emails into relevant categories for different clients. Your employer suspects that some of their machine learning developers are not following best practices and may even be actively sabotaging some of the projects.

Your employer wants you to diagnose what the problems are for projects for seven different clients (listed as questions below, e.g. Q1 OrbitalForge). Each client has provided their own data to construct custom models. Unfortunately, you haven't been able to get access to the code used to construct the machine learning systems. You will have access to the datasets used and the trained models. You should not need to train any models yourself for this work.

For all seven clients, emails are classified into the categories below.

Label	Description
Client & Partner Communications	Customers or partners regarding relationships, projects, and support
HR & Internal Admin	Hiring, policies, benefits, and internal announcements
Operations & Maintenance	Daily operations, issue resolution, maintenance, and operational updates
Product & Engineering	Building, improving, and maintaining products or systems
Sales & Marketing	Sales activities, marketing campaigns, and lead engagement

All available resources for the different client projects are available through Hugging Face Hub and can be located with this link: <https://huggingface.co/TextAsData>. You should use the `transformers` and `datasets` Python packages to access them. Note that all data and clients are fictitious.

Submit your text answers and associated code for each question through the Moodle submission page.

Q1 - OrbitalForge [2 marks]

The original data for this project has been lost. All that remains is the data post-tokenization. But there are no records of which tokenizer was used and the client wants to know. It was likely from the list below. Figure out which tokenizer it was.

- roberta-base
- microsoft/deberta-base
- bert-base-uncased
- distilbert-base-uncased
- bert-base-cased
- allenai/scibert_scivocab_uncased

Q2 - NeuroBloom [3 marks]

This project boasted the development of a brand-new BERT model from scratch including a bespoke tokenizer. What is wrong with the tokenizer?

Q3 - HydroVine [4 marks]

The machine learning model isn't performing well for the client. Diagnose the issue.

Report macro-F1 and accuracy for training/validation/test splits to 3 decimal places.

Q4 - LumenGrid [4 marks]

The client is a little disappointed in the model when deployed as it is not achieving the reported test performance.

Evaluate the performance of the model and decide if anything may be wrong.

Report macro-F1 and accuracy for training/validation/test splits to 3 decimal places.

Q5 - AeroSynth [6 marks]

The client mentions that they've encountered some unusual performance results when they've evaluated the model with the existing dataset. What are the unusual results and what is the cause?

Report macro-F1 and accuracy for training/validation/test splits to 3 decimal places.

Q6 - CryoNest [4 marks]

The client believes there is a problem with the dataset that is causing issues with evaluation. Diagnose what issue has occurred.

Report macro-F1 and accuracy for training/validation/test splits to 3 decimal places.

Q7 - NeonPixel [5 marks]

The client wants to know if the model is ready. Are they able to deploy, and if not, what is the issue?

Report macro-F1 and accuracy for training/validation/test splits to 3 decimal places.

Q8 - SolaraMesh [5 marks]

The client was told that the model got 94% accuracy and is ready to deploy. Was this performance report made in good faith?

Report macro-F1 and accuracy for training/validation/test splits to 3 decimal places.

Q9 - Time Taken [0 marks]

Please write down how many hours you have spent on this coursework. This helps us calibrate coursework difficulty and length for future years.